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SkillSwap MVP Software Requirements Specification

Status: Ready for Review

Version: 1.0

Owner: Group 4

Date: 2026-09-25

Standard: IEEE 830-1998 / ISO/IEC/IEEE 29148:2018 outline

Related documents: SkillSwap-PRD.md , SkillSwap-BRD.md , docs/intent.md

Requirement policy: This SRS contains **confirmed requirements only**. Requirements whose content is governed by an unresolved Open Question (OQ-001–OQ-011 in the BRD) are stated only at their confirmed core; the unconfirmed portions are excluded from the body and listed for traceability in [Appendix A](#). Source status labels from the PRD (Confirmed core , ... open) are preserved in the requirement tables.

1. Introduction

1.1 Purpose

This document specifies the software requirements for the **SkillSwap MVP**, a peer-to-peer online skill marketplace for university students. It defines the functional behavior, external interfaces, performance targets, constraints and attributes that the software must satisfy. Its intended audience is Group 4 Engineering/QA (implementation and test), the Product Owner (scope approval), and reviewers of the SWD392 deliverable.

1.2 Scope

SkillSwap MVP is a responsive, mobile-first **web application** in which:

- Students register, verify their student status, and learn from verified peer Teachers.
- Teachers are verified per skill by domain-expert Verifiers before they may publish classes.
- Classes are booked and paid with an internal **credit wallet** funded through a payment gateway; every financial effect is recorded in an auditable ledger with a 90/10 Teacher/platform split.
- Classes run online in **Jitsi** rooms granted per booking with time-bounded access, with in-platform chat and post-class ratings.

Out of scope (MVP): native mobile apps, AI matching/recommendation, automatic refunds, integration with school student databases, moderation/report tooling beyond ratings, offline classes, languages other than Vietnamese/English (see BRD §4.2).

1.3 Definitions, acronyms and abbreviations

Term	Definition
Learner	A verified student who searches, books and attends classes.

Term	Definition
Teacher	A student approved for at least one skill who publishes and teaches classes.
Verifier	An invited domain expert who reviews Teacher skill evidence.
Administrator	Platform operator who approves student verifications, manages Verifiers and handles exceptions.
Credit	Internal currency unit; initial rate 1 credit = 1,000 VND (BR-005).
Wallet	Per-user ledger-backed balance of available and pending credit.
Booking	A Learner's reservation of a seat in a Class, settled atomically against the wallet ledger.
Settlement	Atomic ledger posting that debits the Learner and splits the price 90% Teacher / 10% platform.
Class	A single online teaching session with skill, schedule, duration, price and capacity.
OQ-xxx	Open Question requiring Product Owner (or delegated) approval before the affected behavior is implemented.
FR / BR / NFR / UC / AC / API / EVT	Functional Requirement / Business Rule / Non-Functional Requirement / Use Case / Acceptance Criterion / API endpoint / Event (webhook) — as numbered in the PRD.
MVP	Minimum Viable Product.
Jitsi	Self-hosted or hosted Jitsi Meet video conferencing service used for classes.

1.4 References

1. [docs/intent.md](#) — SkillSwap intent (problem, outcome, constraints, open questions).
2. [docs/plans/SkillSwap-PRD.md v1.0](#) — Product Requirements Document (FR/BR/NFR/UC/AC/data model/API contract).
3. [docs/plans/SkillSwap-BRD.md v1.0](#) — Business Requirements Document (scope, business rules, A/C/DEP, OQ-001–OQ-011).
4. [docs/plans/2026-09-15-001-feat-skillswap-mvp-requirements-plan.md](#) — requirements plan.
5. [docs/diagrams/c4/generated/structurizr-SystemContext.png](#) — system context diagram (C4 Level 1).
6. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
7. ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.

1.5 Overview

Sections 2 and 3 follow the IEEE 830 structure: overall description, then specific requirements (external interfaces, functional requirements with use cases and state models, performance, design constraints, software system attributes). Appendices A–C carry excluded OQ-governed clauses, the data dictionary, and the acceptance criteria with traceability.

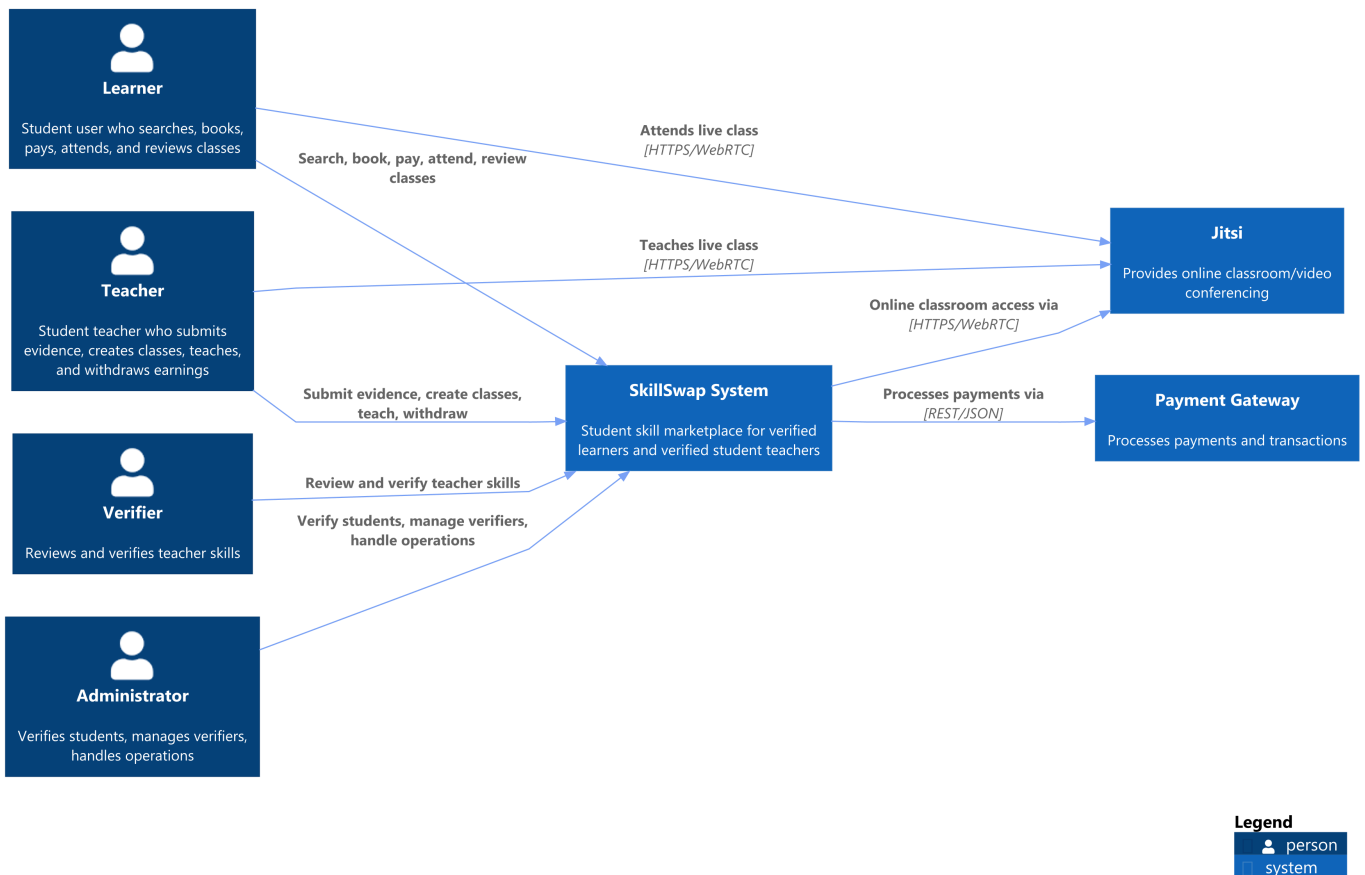
2. Overall description

2.1 Product perspective

SkillSwap is a standalone web application that depends on two external systems: a **Payment Gateway** (real-money top-up and payout) and **Jitsi** (video classrooms). The system context is shown below; container-level detail is in [docs/diagrams/c4/](#).

System Context View: SkillSwap System

System Context diagram for SkillSwap



The software stack is fixed for the MVP: **Next.js (React)** frontend, **NestJS (Node.js)** backend, **PostgreSQL** persistence via **TypeORM/Prisma**. The backend provisions Jitsi rooms and issues short-lived access tokens; Learners and Teachers connect to Jitsi directly via WebRTC, so the backend handles only room metadata and access signaling, never media streams.

2.2 Product functions (summary)

1. **Accounts:** unified Learner/Teacher account registration, login, role view switching (FR-001, FR-017).
2. **Verification:** manual student-status verification by Administrators (FR-002); per-skill Teacher evidence review by matched Verifiers (FR-003, FR-004, FR-018).
3. **Marketplace:** verified Teachers publish classes (FR-005); Learners search/filter classes (FR-006); conditional booking (FR-007).
4. **Finance:** credit top-up via gateway (FR-008), atomic 90/10 settlement (FR-009), wallet balance/history (FR-010), Teacher withdrawal (FR-014).
5. **Learning:** time-bounded Jitsi room access (FR-011), in-booking chat (FR-012), one-shot post-class rating (FR-013).
6. **Operations:** Admin exception dashboard with filterable audit log (FR-015), status notifications (FR-016).

2.3 User characteristics

User	Skill level	Key trait
Learner (persona: Minh, 2nd-year student)	Comfortable with mobile web; low tolerance for complex flows	Books classes on a phone; needs visible account status.

User	Skill level	Key trait
Teacher (persona: Lan, 4th-year student)	Intermediate; motivated by income	Publishes classes, tracks wallet, withdraws earnings.
Verifier (persona: Thầy Hùng, guest lecturer)	Expert in one domain; time-constrained	Reviews queue by domain; approve/reject with structured reasons.
Administrator (persona: Chị Trang, operations)	Power user	Runs verification queues, manages Verifiers, resolves exceptions.

All four roles share one account model: a single account may hold both Learner and Teacher roles with an explicit role view (PRD §6.1).

2.4 Constraints

ID	Constraint	Source
C-001	No school database integration; student verification is fully manual.	BRD §8, Confirmed
C-002	Online classes only (Jitsi).	BRD §8 / BR-007, Confirmed
C-003	Class duration 30 min–3 h; booking at least 24 h in advance.	BRD §8 / BR-003, BR-004, Confirmed
C-004	Initial rate 1 credit = 1,000 VND; changes require approval and versioning.	BRD §8, Confirmed default
C-005	Platform fee 10% per class purchase; 90% allocated to Teacher.	BRD §8 / BR-006, Confirmed
C-006	Mobile-first responsive web; no horizontal scrolling at 360 px/390 px.	PRD NFR-001, Confirmed
C-007	Vietnamese and English only; no mixed language on one screen.	PRD NFR-002, Confirmed
C-008	Stack: Next.js frontend, NestJS backend, PostgreSQL with TypeORM/Prisma.	Engineering decision
C-009	All class communication stays in-platform; contact sharing is prohibited per approved policy.	BR-008, Confirmed (enforcement mechanism deferred — OQ-009)
C-010	MVP has no automated report/account-lock system; ratings + comments surface bad behavior.	intent.md, Confirmed

2.5 Assumptions and dependencies

Assumptions (A-001–A-006, all Unconfirmed — BRD §8): users hold a bank account/e-wallet compatible with the chosen gateway; users accept uploading student documents; enough domain-matched Verifiers exist before Teacher onboarding; users have stable Internet for Jitsi; initial Teacher supply covers 5–10 popular skill categories; Verifier review SLA is 48–72 h.

Dependencies:

ID	Dependency	Status
DEP-001	Payment gateway supporting top-up and payout in Vietnam.	Open (OQ-011)
DEP-002	Jitsi self-hosted or as-a-service with secure domain/JWT.	Open
DEP-003	Initial Verifier headcount.	Open

3. Specific requirements

3.1 External interfaces

3.1.1 User interfaces

- Responsive web UI, mobile-first (360 px, 390 px and desktop viewports); no horizontal scrolling; touch targets sized for touch; WCAG 2.2 AA on main flows (NFR-001, NFR-007).
- Main screens: register/login, Learner home + search, class detail, booking confirmation, student verification form, Teacher registration, Teacher dashboard, class publish/edit, wallet, classroom + chat, rating modal, public Teacher profile, Verifier dashboard, Administrator dashboard, notification center (PRD §8).
- Language: Vietnamese or English per locale, never mixed on one screen (NFR-002).
- Account status (including rejection reasons) must be visible on the Learner home screen with a resubmit path (FR-017).

3.1.2 Hardware interfaces

None. The system is browser-based; the only client hardware assumptions are a camera/microphone for Jitsi participation on the user's own device.

3.1.3 Software interfaces

Interface	Direction	Contract
Payment gateway	Outbound (REST), inbound webhook <code>POST /api/webhooks/payment</code> (EVT-001)	Signature verification, timestamp/replay protection, monotonic status matrix, one application per provider event ID. Specific gateway is OQ-011 (excluded — Appendix A).
Jitsi	Outbound room provisioning + token issuance (<code>POST /api/bookings/{id}/room-token</code> , API-008)	Short-lived authorization for booking participants only; no discoverable public room link. Deployment choice (self-hosted vs. service) is DEP-002.
PostgreSQL	Internal persistence	TypeORM/Prisma data access; financial writes are transactional.
Email service	Outbound	Standard password-recovery flow (PRD §6.1). Provider not fixed at MVP scope level.

3.1.4 Communication interfaces

- All client–server traffic over HTTPS; JSON request/response bodies.
- REST API surface API-001–API-010 and webhook EVT-001 per PRD §10, each with defined auth, payload and error classes.
- Every command with financial effect carries an **idempotency key** and **trace ID**; gateway webhooks carry signature, timestamp and replay nonce.
- Notifications (FR-016) are emitted for: verification outcome, booking confirmed/cancelled, class starting soon, wallet transaction completed, withdrawal outcome. The delivery channel mix (in-app required; push/email optional) is not fixed by an approved requirement and is treated as in-app notification center for MVP scope (see Appendix A, OQ cross-reference).
- Jitsi media flows directly between participants and Jitsi (WebRTC); the application never carries media.

3.2 Functional requirements

Priority follows MoSCoW from the PRD (`Must` / `Should`). Status preserves the PRD's confirmed/open annotation.

3.2.1 Requirement specification

ID	Requirement (shall)	Priority	Source	Status
FR-001	The system shall allow a student to create, sign in to, and manage one account used for the Learner role and Teacher registration.	Must	intent.md	Confirmed
FR-002	The system shall allow a student to declare a school name and upload documents for Administrator approval or rejection with a reason.	Must	intent.md	Confirmed; retention open (OQ-008)
FR-003	The system shall allow a Teacher candidate to submit skills, certificates or capability evidence, with resubmission after rejection.	Must	intent.md	Confirmed core; taxonomy open (OQ-001)
FR-004	The system shall allow an Administrator to invite, assign domains to, suspend or revoke Verifiers; only a matched Verifier may approve a Teacher's evidence.	Must	intent.md	Confirmed
FR-005	Only a verified Teacher shall be able to create and publish an online class with skill, description, schedule, duration, price and approved capacity rules.	Must	intent.md	Confirmed core; price/capacity open (OQ-002, OQ-007)
FR-006	The system shall let a Learner browse and search open classes by class information and Teacher profile.	Must	intent.md	Confirmed core; taxonomy open (OQ-001)
FR-007	The system shall only allow booking a class that has capacity, starts at least 24 hours from now, and lasts 30 minutes to 3 hours.	Must	intent.md	Confirmed core; capacity/refund open (OQ-003, OQ-007)
FR-008	The system shall allow top-up through a payment gateway at the applied rate, with exactly-once callback processing.	Must	intent.md	Confirmed core; gateway open (OQ-011)
FR-009	When a Learner buys a class, the system shall atomically debit the full price and allocate 90% to the Teacher and 10% platform fee.	Must	intent.md	Confirmed split; release timing open (OQ-006)
FR-010	The system shall let a wallet owner view balance and history of top-ups, payments, income, fees and withdrawals.	Must	intent.md	Confirmed core
FR-011	The system shall grant the Learner and Teacher of a booking access to the correct Jitsi room within the permitted time window.	Must	intent.md	Confirmed core
FR-012	The system shall provide in-class chat to booking participants only; outsiders shall be denied and contact sharing handled per approved policy.	Should	intent.md	Confirmed; enforcement open (OQ-009)
FR-013	After class, the system shall allow the Learner to submit exactly one rating/comment per valid booking.	Must	intent.md	Confirmed core; threshold open (OQ-004)
FR-014	The system shall let a Teacher request withdrawal of earnings through the gateway with idempotency and complete ledger records.	Must	intent.md	Confirmed core; policy/gateway open (OQ-005, OQ-011)
FR-015	The system shall provide an Administrator dashboard aggregating the student-verification queue, Verifier queue by domain, dispute/account-lock lists, and an audit log filterable by actor, time and action type.	Must	intent.md	Confirmed
FR-016	The system shall notify users on verification outcomes, booking confirmed/cancelled, class starting soon, completed wallet transactions and withdrawal outcomes.	Should	intent.md	Confirmed
FR-017	The system shall show the Learner their account status and any rejection reason on the main screen.	Must	intent.md	Confirmed
FR-018	The system shall show a Teacher the decision history for each submitted evidence version, not only the latest status.	Should	intent.md	Confirmed

3.2.2 Business rules

ID	Rule	Governs
BR-001	Administrators manually approve student status from school name and uploaded documents, using a standardized rejection-reason list.	FR-002
BR-002	Only a Teacher whose skill was approved by a domain-matched Verifier may open a class for that skill; enforced at API layer, not UI only.	FR-003–FR-005
BR-003	Every class lasts at least 30 minutes and at most 3 hours, including Teacher edits to published classes.	FR-005, FR-007
BR-004	A booking must be created at least 24 hours before class start, computed in the system timezone.	FR-007
BR-005	Initial rate is 1 credit = 1,000 VND; the applied rate is stored per transaction for audit.	FR-008, FR-010, FR-014
BR-006	Platform fee is 10% of class price; the remaining 90% is allocated to the Teacher per settlement state.	FR-009, FR-014
BR-007	MVP supports online classes only.	FR-005, FR-011
BR-008	Class-serving communication happens in-platform; contact sharing is prohibited per approved policy.	FR-012
BR-009	Low-quality profiles are down-ranked per approved threshold, never deleted.	FR-006, FR-013
BR-010	A Teacher may not book or rate their own class.	FR-007, FR-013
BR-011	One account holds at most one effective student verification at a time.	FR-002
BR-012	Credit may not be converted to cash outside the official Teacher withdrawal flow.	FR-014

3.2.3 Use cases

Detailed use cases UC-001–UC-005 are specified in the PRD §11 and restated here with verification-oriented pre/postconditions; UC-006–UC-014 are specified in the Use Case List (§3.2.5).

UC-001 — Verify a student

- Actors: Student (applicant), Administrator. Trigger: user submits student-status verification.
- Main flow: enter school name + upload document → status Pending → Administrator reviews → Approved → status Active.
- Alternate flow: invalid file → Rejected with mandatory reason → user edits and resubmits.
- Postconditions: status, reviewer, timestamp and reason recorded in audit log.

UC-002 — Verify a Teacher skill

- Actors: Teacher candidate, Verifier, Administrator. Trigger: verified student registers to teach a skill.
- Main flow: submit evidence → routed to domain-matched Verifier → review → skill Approved → class publication allowed for that skill.
- Alternate flow: incomplete data, wrong domain, or Reject with reason → Teacher views reason → submits a new version (previous → Superseded).
- Postconditions: only Approved skills may publish classes; full decision history retained (FR-018).

UC-003 — Book and pay for a class

- Actors: Learner, Teacher, Wallet. Trigger: Learner selects an open class.
- Main flow: check eligibility (≥ 24 h, capacity, balance) → hold seat → write booking + ledger entries in one atomic transaction → confirm.
- Alternate flows: insufficient balance / full / duplicate booking / settlement error → no partial success state may exist.
- Postconditions: booking and ledger entries share the same outcome (AC-003–AC-005, AC-010).

UC-004 — Attend and review

- Actors: Learner, Teacher, Jitsi. Trigger: valid booking reaches class start time.
- Main flow: obtain time-bound room token → join room → attend + chat → class ends → booking Completed → Learner submits rating/comment once.
- Alternate flow: non-participant or out-of-window access → denied.
- Postconditions: attendance, incidents and review linked to the booking.

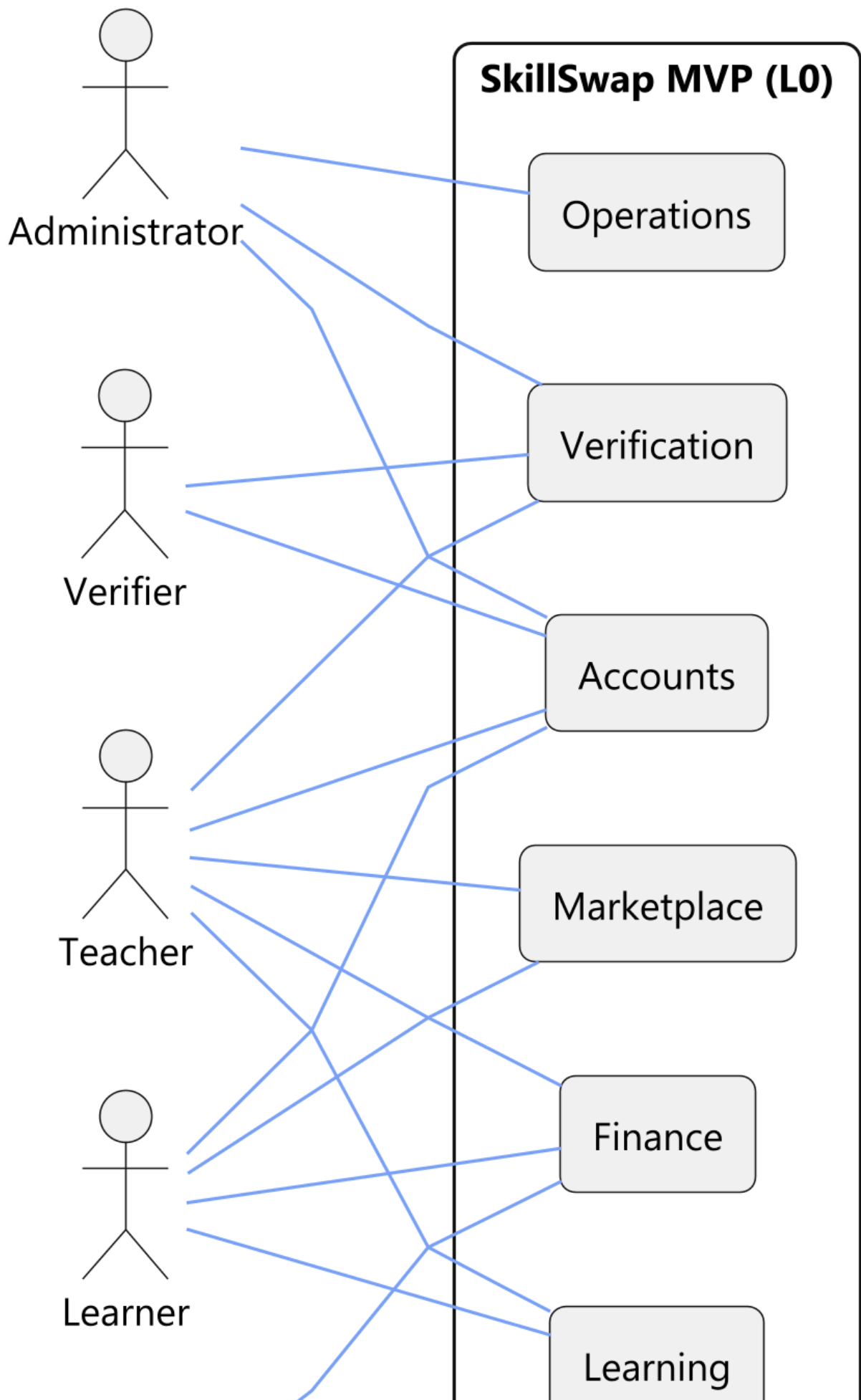
UC-005 — Withdraw earnings

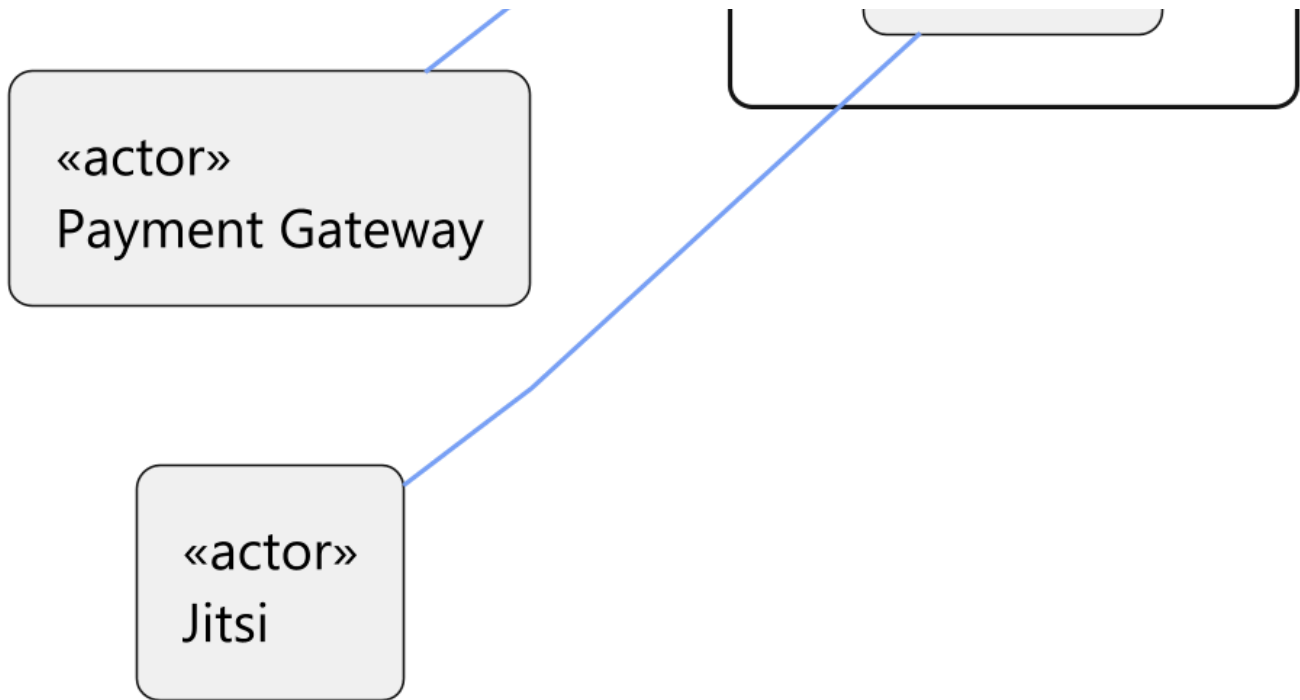
- Actors: Teacher, Payment gateway. Trigger: Teacher requests payout of available balance.
- Main flow: step-up authentication → create withdrawal (Pending) → gateway payout → signed callback → ledger Completed.
- Alternate flow: timeout/failure keeps balances consistent; replayed callback never pays twice.
- Postconditions: payout status and trace ID visible in wallet history (AC-008).

3.2.4 Use case diagrams (level 0 and level 1)

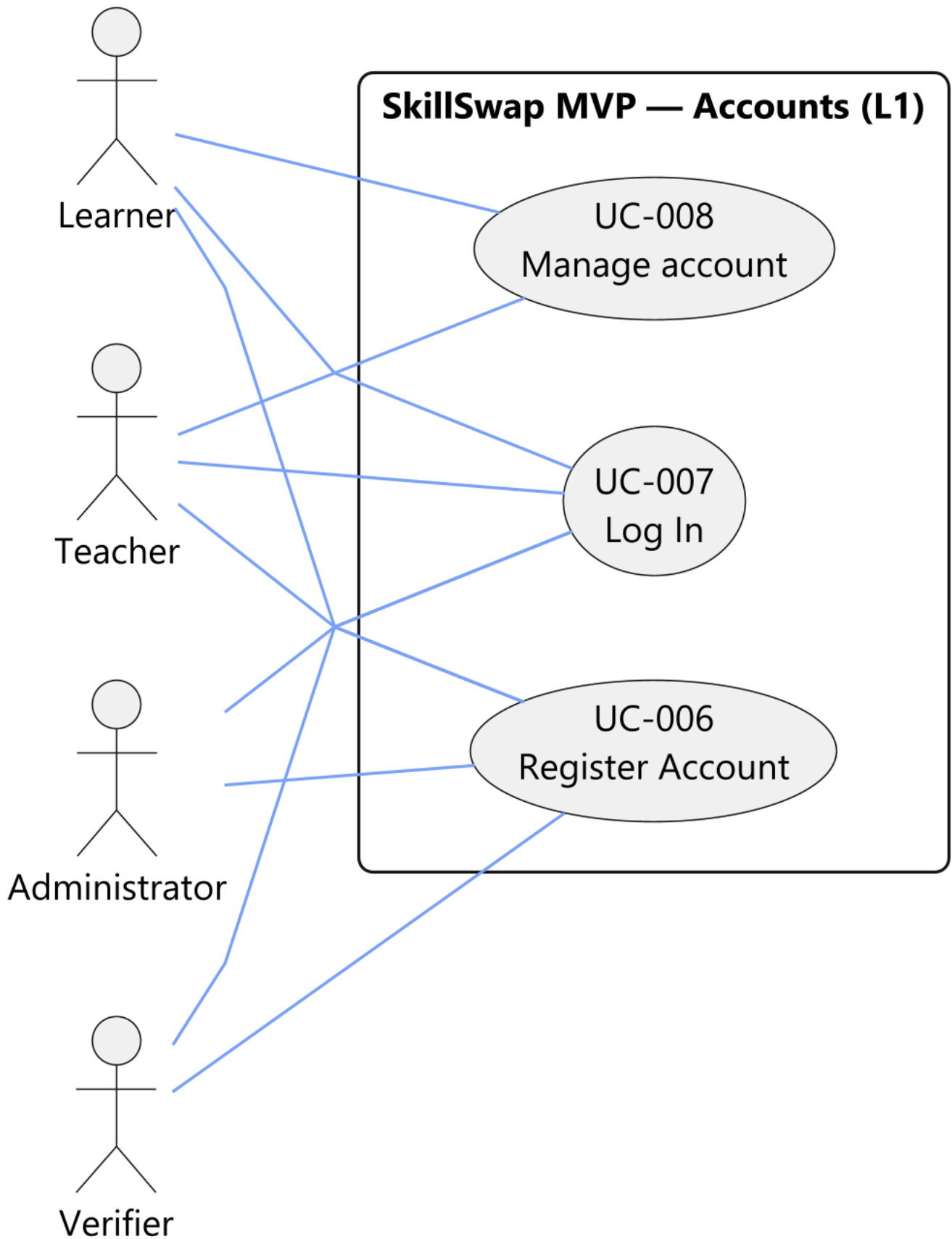
The use case model is leveled into a single **level-0** overview of actors × subsystems and one **level-1** diagram per subsystem. Notation is classic UML: stick-figure actors, elliptical use cases labeled with the immutable code before the name, and unlabeled plain associations connecting actors to use cases only.

Level 0 — actors × subsystems

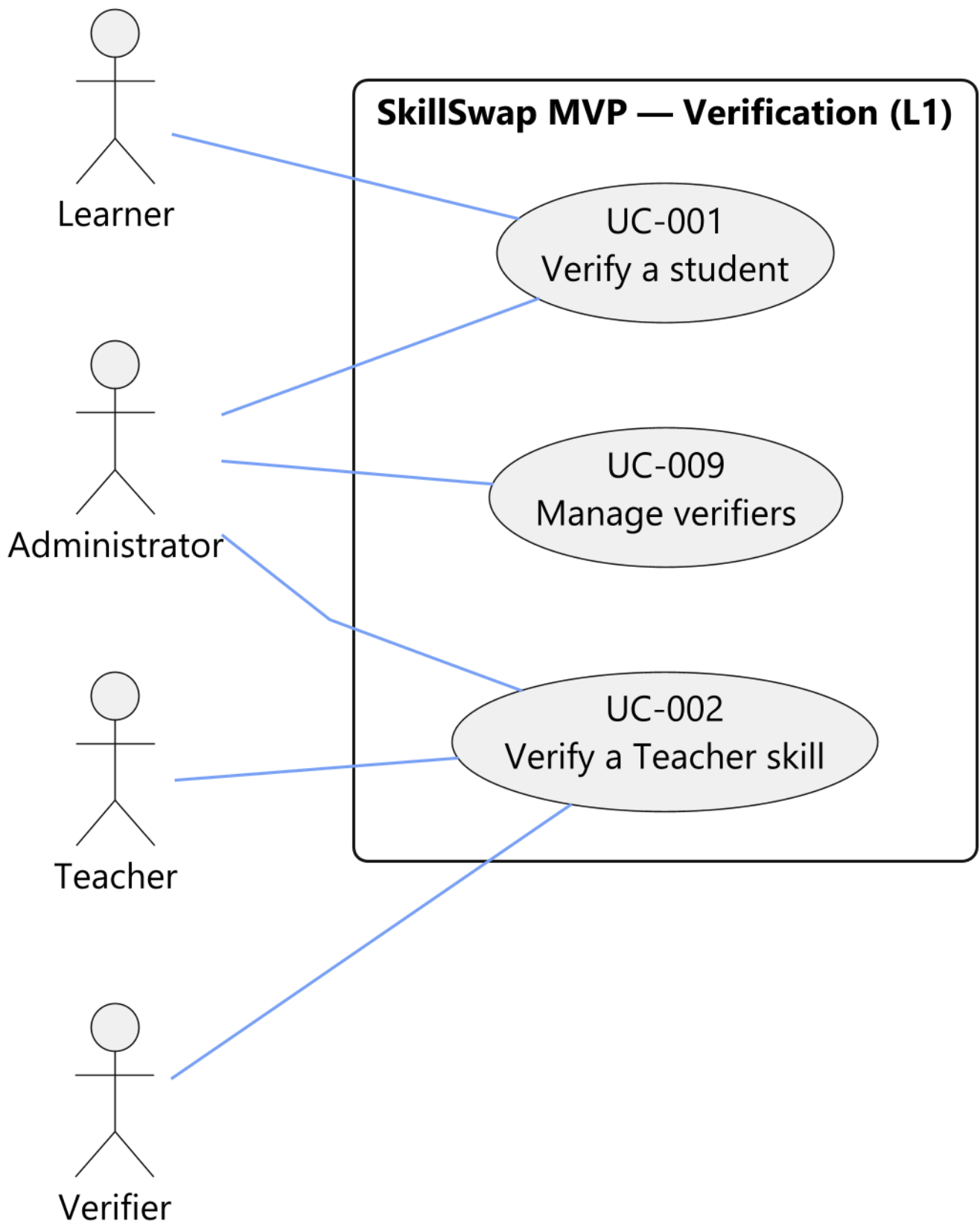




Level 1 — Accounts (UC-006–UC-008)

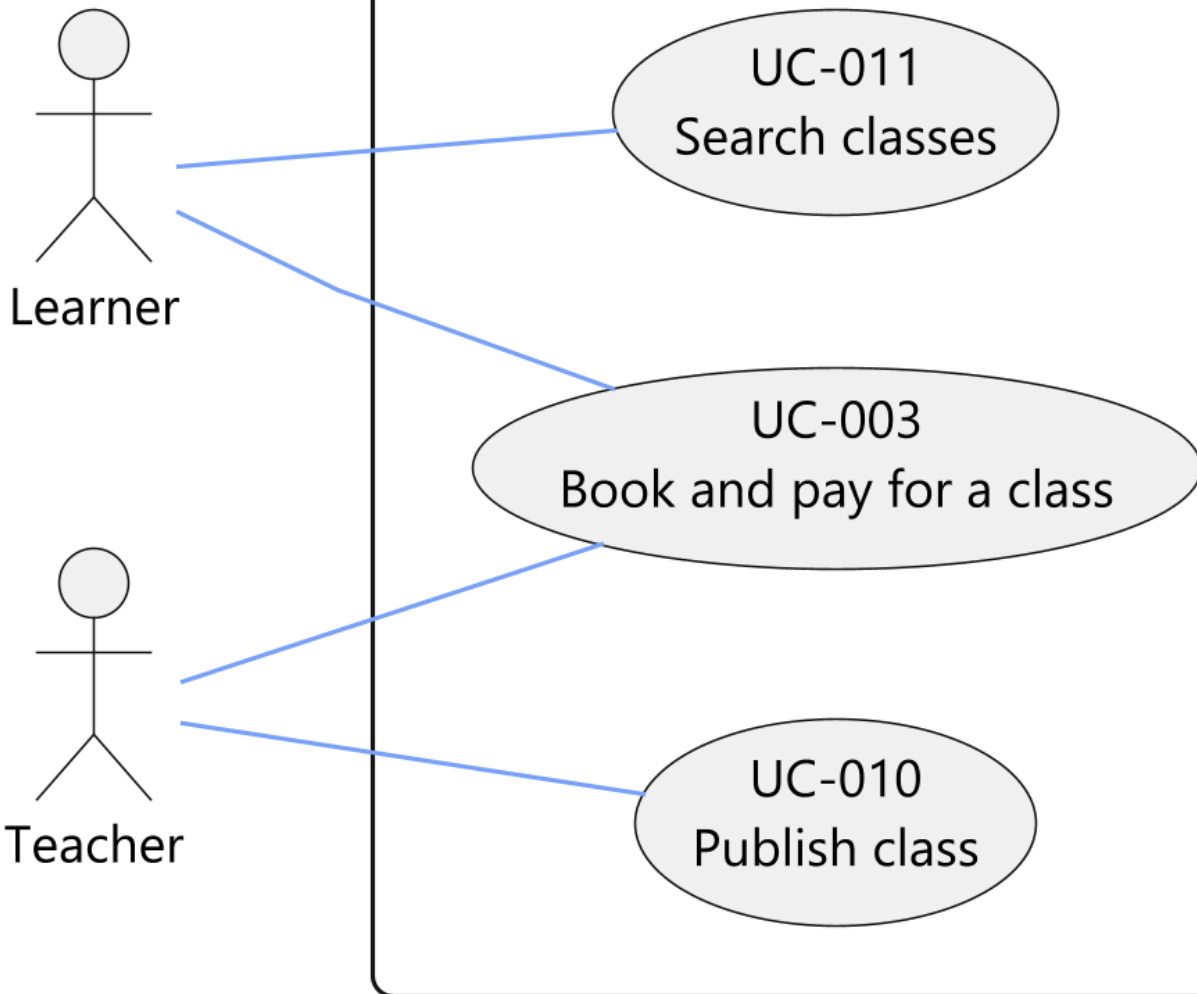


Level 1 — Verification (UC-001, UC-002, UC-009)

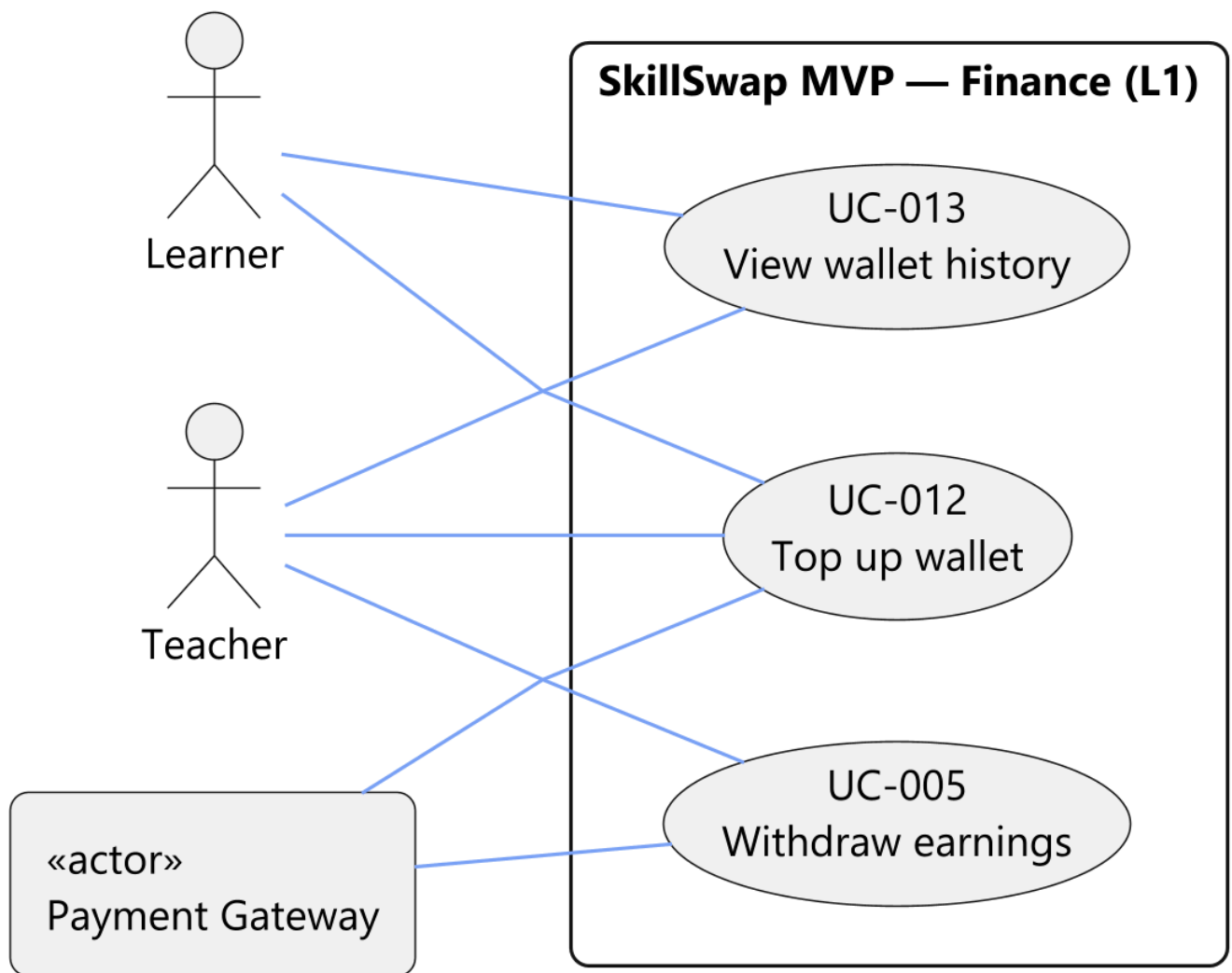


Level 1 — Marketplace (UC-003, UC-010, UC-011)

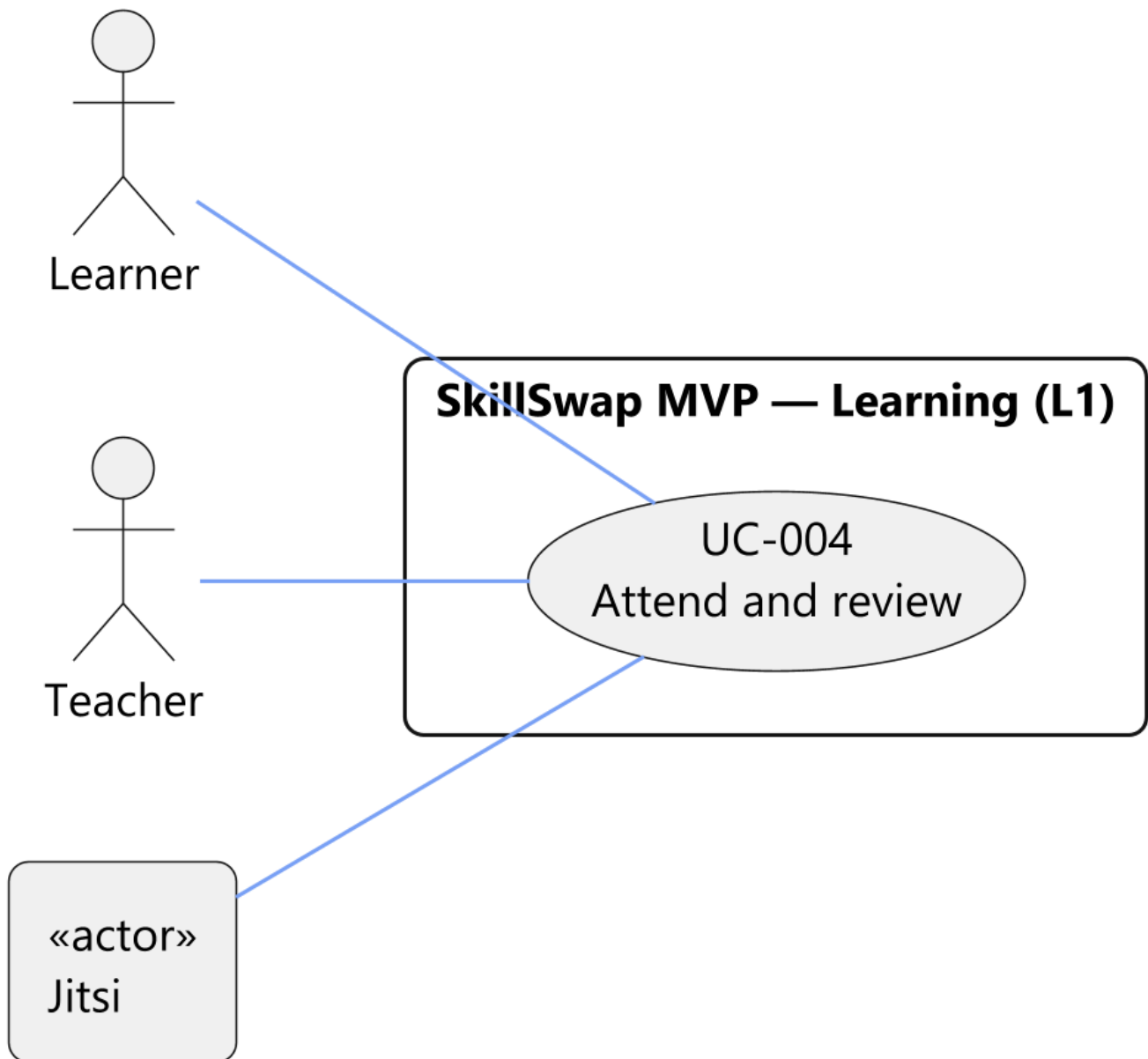
SkillSwap MVP — Marketplace (L1)



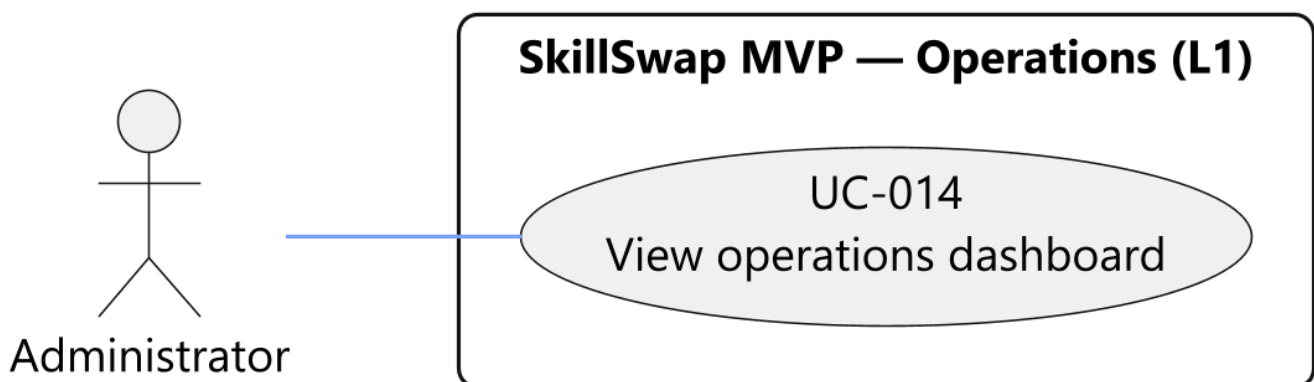
Level 1 — Finance (UC-005, UC-012, UC-013)



Level 1 — Learning (UC-004)



Level 1 — Operations (UC-014)



Reading: every diagram shows the SkillSwap MVP boundary. The four human actors are stick figures outside it; Payment Gateway and Jitsi appear as «actor» boxes because they are external systems. Lines are unlabeled actor–use case associations — there are no include/extend dependencies, because each use case is a standalone user goal and authentication is captured as a Precondition instead of an included Log In use case (§3.2.5). Register Account (UC-

006) and Log In (UC-007) are linked to all four user roles. Every use case carries the identical code and name on its level-1 diagram, in the Use Case List (§3.2.5) and in the traceability matrix (Appendix C.2).

3.2.5 Use Case List

One row per use case appearing on the diagrams, ordered by subsystem then code. Codes are immutable and names match the diagrams exactly.

UC ID	Use Case Name	Primary Actor	Secondary Actor(s)	Subsystem	Description	Precondition	Relationships	Priority
UC-006	Register Account	Learner, Teacher, Verifier, Administrator	—	Accounts	A prospective user creates an account (FR-001); it starts Pending and becomes Active after verification where applicable.	No account exists yet.	—	Must
UC-007	Log In	Learner, Teacher, Verifier, Administrator	—	Accounts	An existing user authenticates and obtains the session required by every other use case.	Account registered (UC-006).	—	Must
UC-008	Manage account	Learner, Teacher	—	Accounts	View and edit the profile, switch the active role view (FR-017), and read account status and rejection reason (FR-001).	Logged in.	—	Must
UC-001	Verify a student	Learner	Administrator	Verification	Submit student-status evidence; an Administrator approves or rejects it, activating the account on approval (FR-002).	Logged in; verification not yet Approved.	—	Must

UC ID	Use Case Name	Primary Actor	Secondary Actor(s)	Subsystem	Description	Precondition	Relationships	Priority
UC-002	Verify a Teacher skill	Teacher	Verifier, Administrator	Verification	Submit skill evidence; a domain-matched Verifier (or the Administrator) decides, enabling class publication for that skill (FR-003, FR-018).	Logged in as a verified student; skill not yet Approved.	—	Must
UC-009	Manage verifiers	Administrator	—	Verification	Invite Verifiers, assign review domains, and suspend or revoke their access (FR-004).	Logged in as Administrator.	—	Must
UC-003	Book and pay for a class	Learner	Teacher	Marketplace	Book an open class and pay from the wallet; booking and ledger entries commit atomically (FR-007, FR-009).	Logged in; class Published with a free seat and sufficient balance.	—	Must
UC-010	Publish class	Teacher	—	Marketplace	Create and publish a class listing for an Approved skill so Learners can discover and book it (FR-005).	Logged in as Teacher; at least one skill Approved.	—	Must
UC-011	Search classes	Learner	—	Marketplace	Discover classes by skill, schedule, price and rating (FR-006).	Registered account.	—	Must
UC-005	Withdraw earnings	Teacher	Payment Gateway	Finance	Withdraw available balance through the gateway; the payout completes exactly once (FR-014).	Logged in as Teacher; available balance > 0.	—	Must

UC ID	Use Case Name	Primary Actor	Secondary Actor(s)	Subsystem	Description	Precondition	Relationships	Priority
UC-012	Top up wallet	Learner, Teacher	Payment Gateway	Finance	Top up wallet credit through the payment gateway (FR-008).	Logged in.	—	Must
UC-013	View wallet history	Learner, Teacher	—	Finance	Review balance, ledger entries and payout status with trace IDs (FR-010).	Logged in; wallet exists.	—	Must
UC-004	Attend and review	Learner	Teacher, Jitsi	Learning	Join the Jitsi room in the permitted window, use in-class chat, and submit one rating after completion (FR-011–FR-013, FR-016).	Confirmed booking within the access window.	—	Must
UC-014	View operations dashboard	Administrator	—	Operations	Review operational metrics and the audit trail (FR-015).	Logged in as Administrator.	—	Must

Notes: the Relationships column is — for every row because no include/extend relationships apply — each use case is a standalone user goal, and authentication is recorded as the Precondition `Logged in` (or `Registered account` / `Logged in as ...`) instead of an included `Log In` use case. Priority follows the governing FR MoSCoW priorities (all Must). The primary/secondary split mirrors the associations drawn in §3.2.4.

3.2.6 State models

Five entity lifecycles are normative. Any transition not listed is forbidden.

1. StudentVerification (FR-002, BR-001, BR-011)

```
stateDiagram-v2
    [*] --> Draft
    Draft --> Pending : submit school name + document
    Pending --> Approved : Administrator approves
    Pending --> Rejected : Administrator rejects with reason
    Rejected --> Draft : user edits and resubmits
    Approved --> Expired : validity period elapses (per retention policy, OQ-008)
    Expired --> Draft : re-verification
    Rejected --> [*]
    note right of Approved
        Only one effective verification
        per account (BR-011)
    end note
```

2. SkillEvidence (FR-003, FR-018, BR-002)

```
stateDiagram-v2
    [*] --> Draft
    Draft --> Pending : submit evidence version N
    Pending --> Approved : matched Verifier approves
    Pending --> Rejected : matched Verifier rejects with reason
    Rejected --> Draft : Teacher submits new version
    Approved --> Superseded : newer version approved
    Approved --> [*] : skill usable to publish classes
    note right of Pending
        Routed only to a Verifier
        matching the skill domain (BR-002)
    end note
```

3. Class (FR-005, FR-007, BR-003)

```
stateDiagram-v2
    [*] --> Draft
    Draft --> Published : verified Teacher publishes (validated schedule/duration/price)
    Published --> Full : last seat booked
    Full --> Published : seat released by allowed cancellation
    Published --> Closed : class concluded
    Full --> Closed : class concluded
    Draft --> Cancelled : Teacher discards
    Published --> Cancelled : Teacher cancels (policy, OQ-003)
    Full --> Cancelled : Teacher cancels (policy, OQ-003)
    note right of Published
        Duration enforced 30 min-3 h (BR-003)
        at every save
    end note
```

4. Booking (FR-007, FR-009, FR-011)

```
stateDiagram-v2
    [*] --> Pending : booking attempt started
    Pending --> Confirmed : atomic settlement succeeded (debit 100%, split 90/10)
    Pending --> [*] : eligibility failed – no side effects
    Confirmed --> Completed : class held, rating window open
    Confirmed --> Cancelled : cancellation per approved policy (OQ-003)
    Completed --> Disputed : dispute raised (manual process)
    Cancelled --> [*]
    Disputed --> [*]
    note right of Confirmed
        No partial success state exists:
        booking and ledger commit or
        roll back together (AC-005)
    end note
```

5. WithdrawalRequest (FR-014, BR-005, BR-012)

```

stateDiagram-v2
    [*] --> Pending : Teacher requests (step-up auth)
    Pending --> Processing : gateway payout accepted
    Processing --> Completed : signed callback succeeded (once)
    Processing --> Failed : gateway timeout / failure
    Failed --> [*] : balance restored immediately
    Completed --> Reversed : valid reversal event only
    note right of Processing
        Replayed provider_event_id is
        acknowledged, never re-applied
        (AC-008, AC-009)
    end note

```

3.3 Performance requirements

ID	Requirement	Measurement	Priority
NFR-011	Search results shall be returned promptly at MVP data scale.	Basic load test (target latency to be fixed by Product Owner — OQ-010 excluded, see Appendix A).	Should
NFR-012	Uptime appropriate to the pilot stage shall be maintained.	Uptime monitoring.	Should
NFR-003	Concurrent or retried financial requests shall produce exactly one balanced financial outcome and never a negative balance.	Idempotency, concurrency and reconciliation tests.	Must
NFR-006	Validation/gateway/Jitsi failures shall preserve recoverable user input and show actionable errors.	Negative-path tests for forms, gateway and Jitsi.	Should

3.4 Design constraints

1. **Architecture:** Next.js frontend, NestJS backend, PostgreSQL via TypeORM/Prisma (C-008); Jitsi room provisioning and token issuance by backend; media flows directly over WebRTC.
2. **Atomicity:** all booking/settlement writes occur in a single database transaction; no partial financial success (NFR-003, AC-005).
3. **Idempotency:** every financially effective command requires an idempotency key and trace ID (PRD §10 integration rules).
4. **Authorization:** server-side role/ownership checks on every endpoint; Administrator/Verifier require MFA; payouts and payout-destination changes require step-up authentication (NFR-009).
5. **Mobile-first:** responsive layouts validated at 360 px, 390 px and desktop (NFR-001, C-006).
6. **Localization:** Vietnamese/English with locale-complete routes (NFR-002, C-007).
7. **Legal:** personal-data handling for identity documents must follow applicable data-protection rules; funds flow exclusively through a licensed gateway; Terms of Service must state Teacher tax responsibility and the under-18 policy (BRD §12).

3.5 Software system attributes

Reliability / availability (NFR-003, NFR-006, NFR-012): no inconsistent financial states under retry/concurrency; pilot-grade uptime with monitoring.

Security (NFR-004, NFR-008, NFR-009, NFR-010, PRD §13):

- Only the owner and authorized reviewers may access verification documents.

- Uploads validated by extension, MIME and file signature, size-limited (10 MB images/PDF per PRD §6.2), malware-scanned, stored outside executable paths, previewed in a sandboxed frame.
- Webhooks authenticated by signature, timestamp and replay nonce; secrets stored outside source control, separated per environment, rotatable.
- Chat/rating content treated as untrusted, context-encoded, rendered under a Content Security Policy (stored-XSS tests).
- Jitsi access uses short-lived tokens for booking participants inside the permitted window only (AC-006).
- Privileged actions, verification decisions, role changes, financial transitions and payout-destination changes are audited with actor, timestamp and trace ID (FR-015, NFR-005).

Maintainability: requirements, acceptance criteria and tests keyed by stable IDs (FR/BR/NFR/UC/AC) enabling the traceability matrix in Appendix C.

Portability: responsive web only; browser support covers current mobile and desktop browsers (BRD §4.1).

Accessibility (NFR-007): WCAG 2.2 AA on main flows — keyboard navigation, visible focus, screen-reader labels.

Privacy (NFR-004, NFR-005): sensitive data (student IDs, certificates, payout destinations, financial history) encrypted in transit and at rest and excluded from logs; top-up/settlement/payout flows carry an end-to-end trace ID without secret leakage.

3.6 Other requirements

- **Data handling (PRD §9):** access is server-authorized by role and ownership; sensitive files encrypted in transit/at rest, stored outside executable paths, never written to application logs.
- **Auditability (NFR-005):** trace one sample transaction through gateway → API → ledger.
- **Testing evidence:** acceptance criteria AC-001–AC-010 (Appendix C) must pass before the corresponding increment is Delivery Ready (PRD §14).

3.7 Open items blocking implementation

Implementation of the clauses listed in Appendix A is blocked until the corresponding Open Question is approved (decision policy: **block until resolved**). Affected increments per PRD §14: Increment 3 (OQ-001, OQ-007), Increment 4 (OQ-002, OQ-003, OQ-005, OQ-006, OQ-011), storage/lifecycle design (OQ-008), Increment 5 (OQ-004, OQ-009), pilot planning (OQ-010).

Appendix A — Excluded open-question-governed clauses

Confirmed cores of these requirements appear in §3.2.1; the following **dependent clauses are intentionally excluded** from this SRS until the governing OQ is approved (BRD §11).

OQ	Decision required	Owner	Affects	Excluded clause (not specified in this SRS)
OQ-001	Skill taxonomy structure and search granularity.	Product Owner	FR-003, FR-005, FR-006	Exact category tree, free-text/tag hybrid, search index behavior.
OQ-002	Teacher self-pricing vs. price bands vs. hybrid with floor/cap.	Product Owner	FR-005, FR-007–FR-009	Price rules, minimum/maximum price validation.
OQ-003	Cancellation, no-show, dispute, manual refund and Admin enforcement workflow.	Product Owner + Administrator	FR-007, FR-009–FR-011, FR-013	Cancellation paths in Class/Booking state models, refund handling, dispute procedure.
OQ-004	Rating threshold, minimum review count, visibility recovery.	Product Owner	FR-006, FR-013	Down-ranking threshold mechanics (BR-009 value).

OQ	Decision required	Owner	Affects	Excluded clause (not specified in this SRS)
OQ-005	Withdrawal conditions, limits, fees, processing time.	Product Owner + Finance	FR-010, FR-014	Minimum/maximum payout, fees, SLA in WithdrawalRequest model.
OQ-006	When the Teacher's 90% moves pending → available; chargeback/reversal handling.	Product Owner + Finance	FR-008, FR-009, FR-014	Settlement release timing (value behind Completed → Reversed policy).
OQ-007	1-1 vs. group classes; capacity and seat hold/release rules.	Product Owner	FR-005, FR-007	Capacity semantics in FR-007, seat-hold logic around Full ⇌ Published.
OQ-008	Verification validity period, document retention/deletion, expiry cascade.	Product Owner + Security	FR-002–FR-005, FR-007, FR-011, FR-014, NFR-004	Retention duration, Approved → Expired timing, deletion/anonymization rules.
OQ-009	Chat handling of detected personal contact info: block / warn+log / log only.	Product Owner	FR-012	Enforcement mechanism behind BR-008.
OQ-010	Target values for success metrics and pilot scope.	Product Owner	Product rollout	Quantified performance/availability targets for NFR-011/NFR-012.
OQ-011	Payment gateway selection (top-up, payout, webhook, reconciliation, reversal).	Engineering + Finance	FR-008, FR-014	Gateway-specific integration contract in §3.1.3.

Appendix B — Data dictionary

Derived from PRD §9. Access is server-authorized by role and ownership; sensitive files are encrypted in transit and at rest.

Entity	Key fields	Relationships	Lifecycle	Classification	Owner
User	id, roles, school_name, verification_status	Has verification, wallet, classes/bookings	Pending → Active → Suspended	PII	User/Admin
StudentVerification	id, user_id, document_ref, status, reviewer_id, reason, expires_at	Belongs to User	Draft → Pending → Approved/Rejected/Expired	Sensitive identity document	Administrator
SkillEvidence	id, teacher_id, skill_id, document_ref, version, status	Belongs to Teacher and Skill	Draft → Pending → Approved/Rejected/Superseded	Sensitive credential	Teacher/Verifier
VerifierAssignment	id, verifier_id, domain, status, assigned_by	Links Verifier to review domain	Invited → Active → Suspended/Revoked	Internal access-control data	Administrator
Class	id, teacher_id, skill_id, schedule, duration, price, capacity, status	Has Bookings	Draft → Published → Full/Closed/Cancelled	Public marketplace data	Teacher
Booking	id, class_id, learner_id, status, settlement_id	Links Class, Learner and ledger	Pending → Confirmed → Completed/Cancelled/Disputed	Private transaction data	Learner/Teacher
Wallet	id, user_id, available_balance, pending_balance	Has ledger entries	Active → Restricted/Closed	Financial	User/Platform

Entity	Key fields	Relationships	Lifecycle	Classification	Owner
LedgerTransaction	id, wallet_id, type, amount, rate, status, trace_id, idempotency_key	Links top-up, booking or withdrawal	Pending → Posted/Reversed/Failed	Sensitive financial	Platform
Message	id, class_id, sender_id, body, created_at	Belongs to class conversation	Active → Retained/Deleted	Private communication	Participants/Platform
Rating	id, booking_id, learner_id, score, comment	One policy-valid rating per Booking	Published → Hidden/Updated by policy	Public content + private provenance	Learner/Platform
WithdrawalRequest	id, teacher_id, amount, destination_ref, status, trace_id	Produces ledger entries	Pending → Processing → Completed/Failed/Reversed	Sensitive financial	Teacher/Platform

Appendix C — Acceptance criteria and traceability

C.1 Acceptance criteria (AC-001–AC-010)

- **AC-001 / FR-002:** Given a school name and a valid file, when the user submits, the status is Pending and an Administrator can review it.
- **AC-002 / FR-003–FR-005:** Given an unapproved skill, when a Teacher attempts to publish a class, the system refuses and directs them to complete verification.
- **AC-003 / FR-007:** Given a class starting in under 24 hours, when a Learner books, no booking is created and no credit is deducted.
- **AC-004 / FR-009:** Given a 100-credit class and sufficient balance, when purchased, the ledger debits 100, allocates 90 to the Teacher pending settlement, records the 10 fee, and creates exactly one booking.
- **AC-005 / FR-009:** Given a settlement step failure, the transaction ends with no booking or balance in a partially successful state.
- **AC-006 / FR-011:** Given a user outside the booking or outside the access window, when they open the room, access is denied.
- **AC-007 / FR-013:** Given a completed class and an eligible Learner, when a valid rating/comment is submitted, the review links to the booking and cannot be duplicated.
- **AC-008 / FR-014:** Given a valid payout callback replayed, the withdrawal completes exactly once.
- **AC-009 / EVT-001:** Given a valid callback older than the processed gateway state, the system acknowledges it but does not lower the status.
- **AC-010 / FR-007:** Given a class with one seat left, when two Learners book near-simultaneously, exactly one booking confirms; the other receives a full-class error with no credit deducted.

C.2 Traceability matrix

Business goal	Requirements	Use case	API / data / NFR
Trusted student identity	FR-001, FR-002, FR-017	UC-001, UC-006–UC-008	API-001, User, StudentVerification, NFR-004/008/009
Trusted Teacher capability	FR-003–FR-005, FR-018	UC-002, UC-009, UC-010	API-002–API-004, SkillEvidence, VerifierAssignment, NFR-004/009

Business goal	Requirements	Use case	API / data / NFR
Discover and book online learning	FR-005–FR-007	UC-003, UC-011	API-004–API-005, Class, Booking, NFR-001/006/007
Traceable wallet settlement	FR-008–FR-010, FR-014	UC-003, UC-005, UC-012, UC-013	API-005–API-007, EVT-001, Wallet/Ledger/Withdrawal, NFR-003/005/009
Secure class participation	FR-011–FR-013, FR-016	UC-004	API-008–API-010, Message/Rating, NFR-006/007/010
Platform operations	FR-015	UC-014	Audit log, NFR-005

C.3 Diagrams index

Use case diagrams are authored in PlantUML (`docs/diagrams/usecase/srs-use-case-*.puml`) and referenced as PNGs, because GitHub does not render PlantUML natively; state diagrams and the C4 context embed Mermaid/PNG as noted.

Diagram	Section	Syntax	Rendered PNG
Use case level 0 (actors × subsystems)	§3.2.4	PlantUML <code>usecase</code> / <code>rectangle</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-10.png</code>
Use case level 1 — Accounts (UC-006–UC-008)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-accounts.png</code>
Use case level 1 — Verification (UC-001, UC-002, UC-009)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-verification.png</code>
Use case level 1 — Marketplace (UC-003, UC-010, UC-011)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-marketplace.png</code>
Use case level 1 — Finance (UC-005, UC-012, UC-013)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-finance.png</code>
Use case level 1 — Learning (UC-004)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-learning.png</code>
Use case level 1 — Operations (UC-014)	§3.2.4	PlantUML <code>usecase</code> (straight links)	<code>docs/diagrams/usecase/srs-use-case-operations.png</code>
StudentVerification state machine	§3.2.6	Mermaid <code>stateDiagram-v2</code>	<code>docs/diagrams/srs-state-student-verification.png</code>
SkillEvidence state machine	§3.2.6	Mermaid <code>stateDiagram-v2</code>	<code>docs/diagrams/srs-state-skill-evidence.png</code>
Class state machine	§3.2.6	Mermaid <code>stateDiagram-v2</code>	<code>docs/diagrams/srs-state-class.png</code>
Booking state machine	§3.2.6	Mermaid <code>stateDiagram-v2</code>	<code>docs/diagrams/srs-state-booking.png</code>
WithdrawalRequest state machine	§3.2.6	Mermaid <code>stateDiagram-v2</code>	<code>docs/diagrams/srs-state-withdrawal-request.png</code>
System context (C4 L1)	§2.1	Embedded PNG from <code>docs/diagrams/c4/generated/</code>	<code>docs/diagrams/c4/generated/structurizr-SystemContext.png</code>